Contents

Part 1- The Four Pillars of Computational Thinking

Task 1- Decomposition (Break it Down)

1. Get member name from staff
2. Get equipment name used
3. Get number of minutes used
4. Calculate usage score
5. Display the result
6. Repeat or end program

Task 2- Pattern Recognition (Identify Trends)

Usage Score = Minutes × 1.5

Task 3- Abstraction (Filter the Essentials)

Essential Variables	Irrelevant Variables (Not

(Important)	Needed)
member_name	member’s outfit color
equipment_name	member’s age
minutes_used	gym music playing

Task 4- Algorithm Design (Step-by-Step Logic)

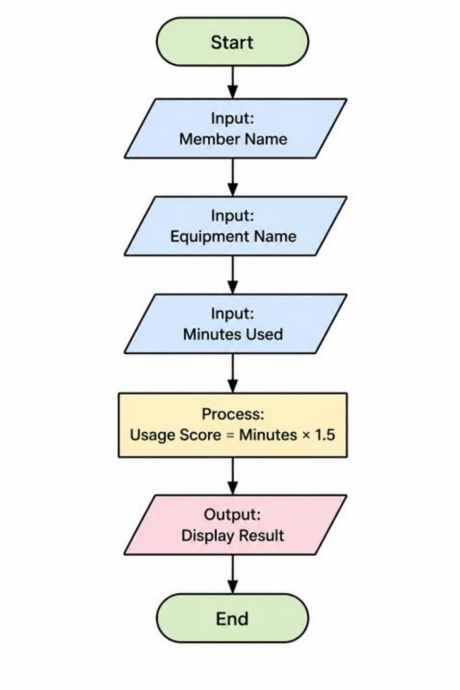
- Start the program
- Ask staff to enter member name
- Ask for equipment name
- Ask for minutes used
- Convert minutes into a number
- Calculate usage score (minutes × 1.5)
- Display the result
- End program

Part 2- Behavioral Design

Task 2.1 - IPO Model (Input-Process-Output)

- Input
  - Member name
  - Equipment name
  - Minutes used
- Process
  - Multiply minutes by 1.5
- Output
  - Member name
  - Equipment name
  - Usage score

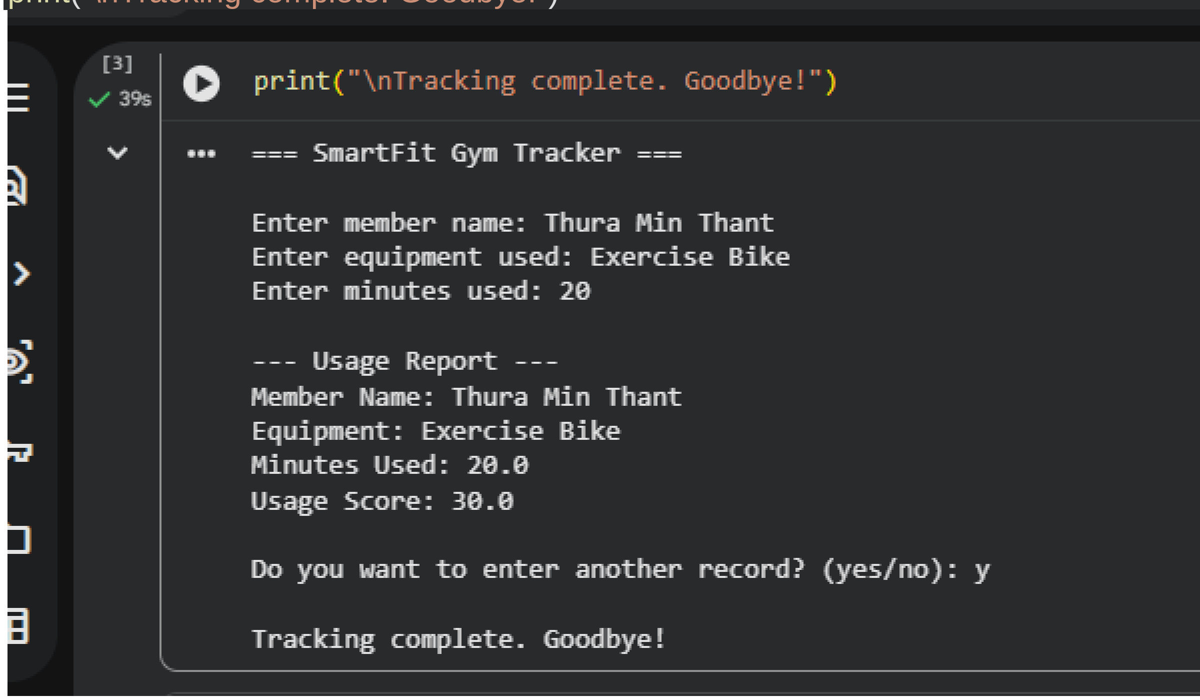
Task 2.2: The Logic Flowchart Guide



Part 3: Python Implementation

```python
# SmartFit Gym - Daily Member Equipment Tracker

print("=== SmartFit Gym Tracker ===\n")

while True:
    # Step 1: Get member name
    member_name = input("Enter member name: ")

    # Step 2: Get equipment name
    equipment_name = input("Enter equipment used: ")

    # Step 3: Get minutes used (with error handling)
    try:
        minutes_used = float(input("Enter minutes used: "))
    except:
        print("Invalid input! Please enter a number.\n")
        continue  # restart loop

    # Step 4: Calculate usage score
    usage_score = minutes_used * 1.5

    # Step 5: Display result
```

```
    print("\n--- Usage Report ---")
    print("Member Name:", member_name)
    print("Equipment:", equipment_name)
    print("Minutes Used:", minutes_used)
    print("Usage Score:", usage_score)

    # Step 6: Ask to continue
    choice = input("\nDo you want to enter another record? (yes/no): ").lower()

    if choice != "yes":
        break

print("\nTracking complete. Goodbye!")
```



```
[3]
39s
print("\nTracking complete. Goodbye!")

=== SmartFit Gym Tracker ===

Enter member name: Thura Min Thant
Enter equipment used: Exercise Bike
Enter minutes used: 20

--- Usage Report ---
Member Name: Thura Min Thant
Equipment: Exercise Bike
Minutes Used: 20.0
Usage Score: 30.0

Do you want to enter another record? (yes/no): y

Tracking complete. Goodbye!
```

Output